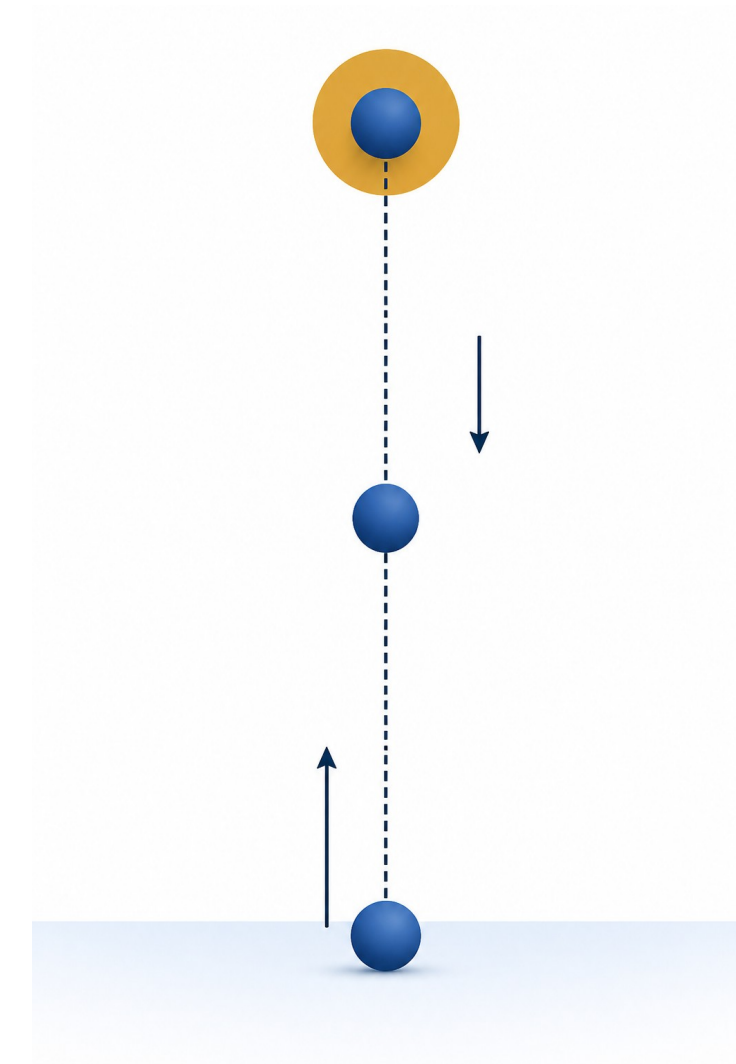


From a Physical Target to a Checkable Root

Week 5 | Projectile range as a checkable residual



physical model

residual

bracket

update

validation

Turn the target condition into a residual

The unknown is an angle; the mismatch is a distance

Model first, then define the mismatch

$$R(\theta) = \frac{v_0^2}{g} \sin(2\theta), \quad v_0 = 20 \text{ m s}^{-1}, \quad g = 9.81 \text{ m s}^{-2}$$
$$R(\theta) = 35 \text{ m} \quad \implies \quad f(\theta) = R(\theta) - 35 \text{ m} = 0$$

unknown → theta in degree

residual → f(theta) in m



**root when the
mismatch is zero**

Target condition

Choose an angle that makes
the model range equal the
target.

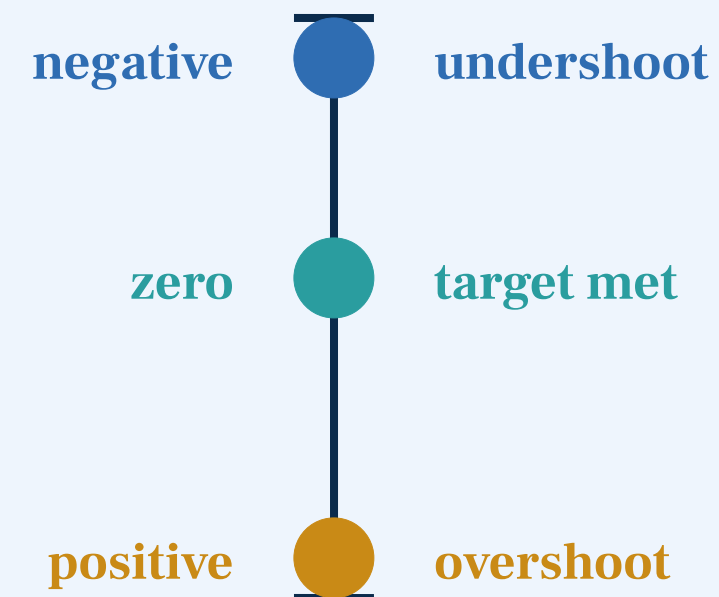
35 m

A root is an input that makes the model–target mismatch vanish.

Read the residual sign as physics

The sign tells us which side of the target the model is on

One sign, one physical meaning



The residual is a signed model error

$$\begin{aligned} f(\theta) < 0 &\implies \text{model range is shorter: undershoot} \\ f(\theta) = 0 &\implies \text{model range equals target: root} \\ f(\theta) > 0 &\implies \text{model range is longer: overshoot} \end{aligned}$$

A negative value tells us which side to keep.

A positive value tells us the target has been passed.

Read the sign before you read the iteration count.

Find the zero crossing before coding

A sign change selects the low-angle bracket

Defend the bracket

$$f(20^\circ) = -8.7905 \text{ m}$$

$$f(40^\circ) = +5.1553 \text{ m}$$

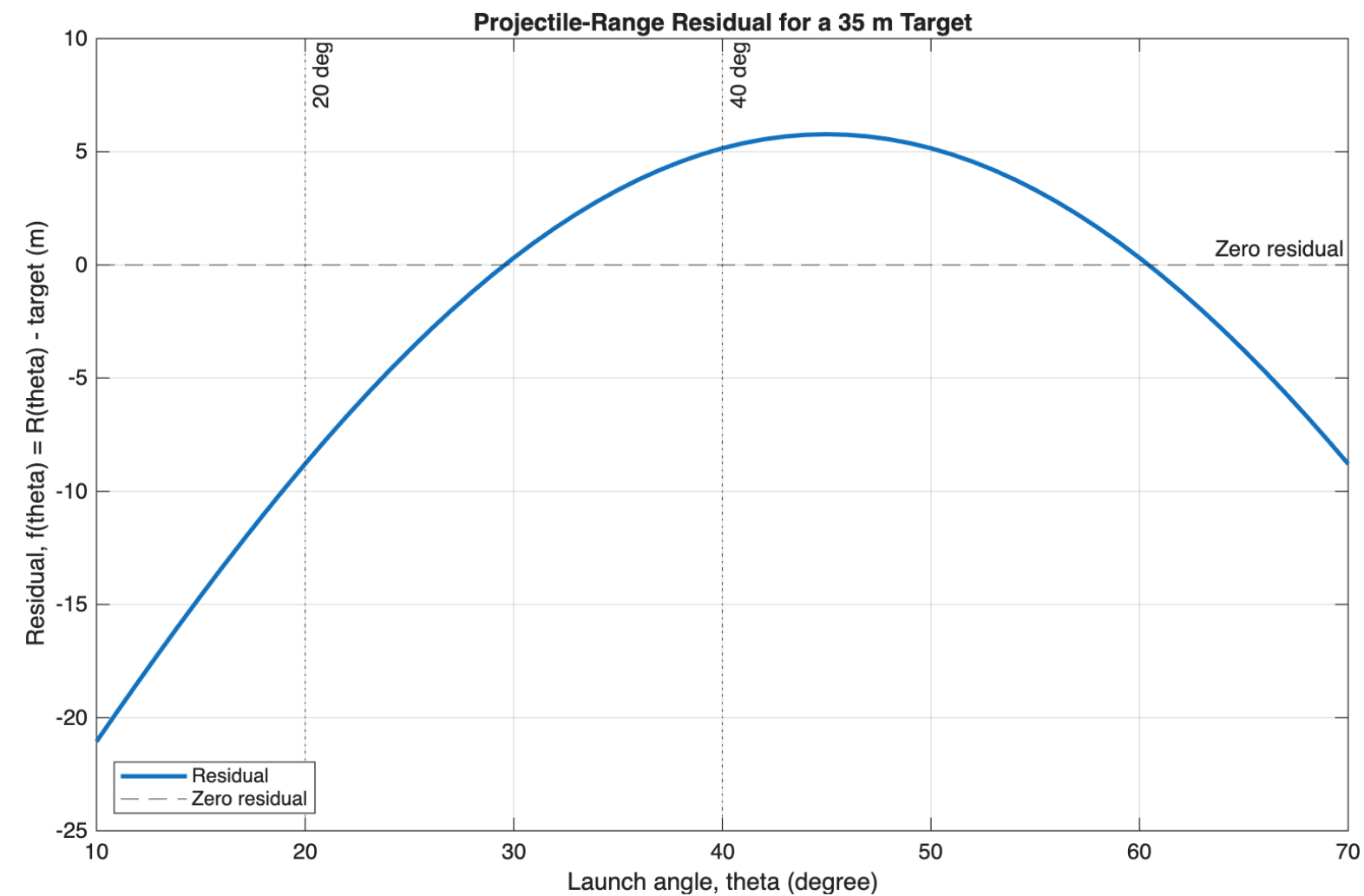


20
degree



40
degree

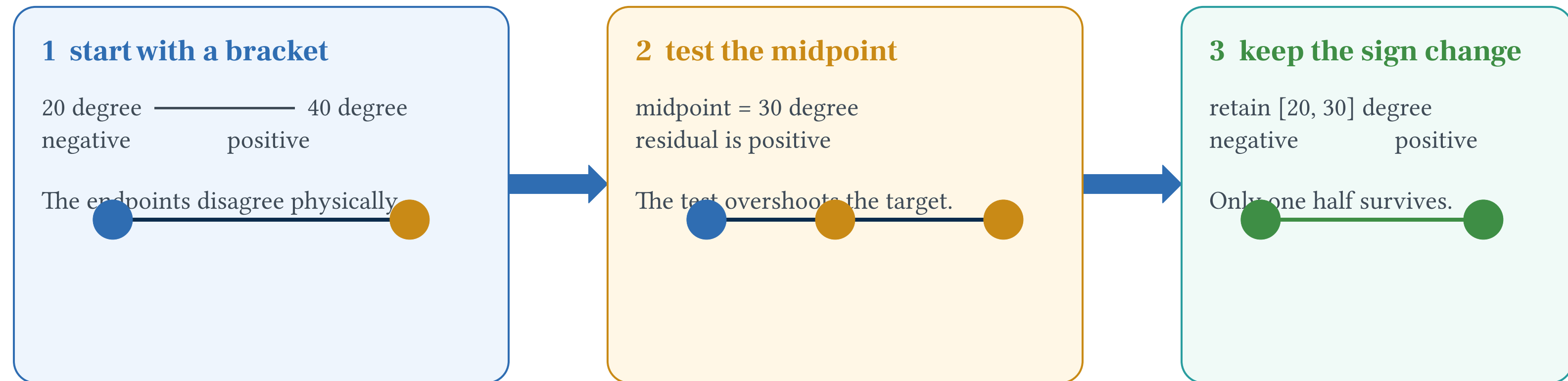
opposite signs → keep [20, 40] degree



A sign-changing bracket is evidence for the next update.

Bisection keeps a sign change alive

Each update preserves the evidence that a zero crossing lies inside



The midpoint is a test; the sign-changing bracket is the safety structure.

Trace the first four bisection updates

Read each row as midpoint → residual → retained half

iteration	bracket (degree)	midpoint (degree)	residual (m)	keep
1	[20, 40]	30	+0.31194	[20, 30]
2	[20, 30]	25	−3.7648	[25, 30]
3	[25, 30]	27.5	−1.5993	[27.5, 30]
4	[27.5, 30]	28.75	−0.61095	[28.75, 30]

The interval shrinks because the sign-changing evidence is preserved.

Midpoint → residual → retained half: that is the bisection pattern.

Read the bisection scaffold, line by line

The MATLAB loop mirrors the physical update logic

Plain-language algorithm

- 1 Copy the bracket**
Start with left and right angles.
- 2 Compute midpoint**
Test the centre of the interval.
- 3 Evaluate residual**
Measure the model mismatch in metres.
- 4 Stop or update**
Keep the half whose signs disagree.

MATLAB scaffold

```
bis_left_deg = left_deg;  
bis_right_deg = right_deg;  
for iteration = 1:max_iterations  
    mid_deg = (bis_left_deg + bis_right_deg)/2;  
    f_mid_m = (v0_mps^2/g_mps2)*sind(2*mid_deg) - target_m;  
    if abs(f_mid_m) < tolerance_m  
        break  
    end  
    f_bis_left_m = (v0_mps^2/g_mps2)*sind(2*bis_left_deg) - target_m;  
    if f_bis_left_m*f_mid_m < 0  
        bis_right_deg = mid_deg;  
    else  
        bis_left_deg = mid_deg;  
    end  
end  
bisection_root_deg = mid_deg;
```

Each line mirrors a physical question already answered in the trace.

Stop once, then substitute back

The stopping rule is only the first half of the check

One Core stopping rule

$$\text{abs(residual_m)} \cdot 1\text{e-}3 \text{ m}$$

Bisection reaches the low-angle root after 13 recorded iterations.

Then substitute the root into the original model.

Substitution check

$$\begin{aligned}\theta_* &= 29.5679 \text{ degree} \\ R(\theta_*) &= 35.0004 \text{ m} \\ f(\theta_*) &= R(\theta_*) - 35 \text{ m} = +4.1164 \times 10^{-4} \text{ m}\end{aligned}$$

check	result	interpretation
root angle	29.5679 degree	low-angle launch angle
model range	35.0004 m	target condition is met
residual	$+4.1164 \times 10^{-4} \text{ m}$	below 10^{-3} m tolerance

The model range is 35.0004 m, within the supplied 1e-3 m tolerance of the target.

Newton follows the local tangent

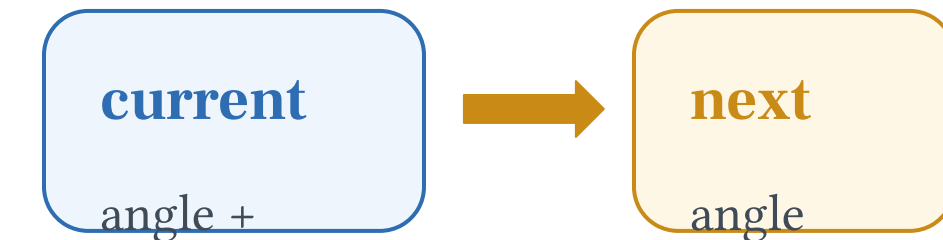
A local slope predicts the next angle

The update uses residual ÷ slope

$$\text{new angle} = \text{current angle} - \frac{\text{current residual}}{\text{current residual slope}}$$
$$\frac{df}{d\theta} = \frac{v_0^2}{g} \cos(2\theta) \frac{2\pi}{180} \quad (\text{metres per degree})$$

Because theta is stored in degree, slope is in m per degree.

Local picture



Fast when the local slope is useful; still validate by residual.

The degree conversion keeps slope units compatible with the update.

Trace Newton from 25 degree

The table records the quantities needed to explain each update

evaluation	theta (degree)	residual (m)	slope (m/degree)	next angle (degree)
1	25	-3.7648	0.91488	29.115
2	29.115	-0.33461	0.74939	29.562
3	29.562	-0.0042242	0.73044	29.567
4	29.567	-7.1308e-7	0.73019	stop

At evaluation 4, the same residual stopping rule is satisfied.

Current angle → residual → slope → next angle: make the update explainable.

Working exposure: same root, different route

Both methods carry evidence, but the evidence is different

Bisection

update clue
endpoint signs + midpoint

recorded cost
13 iterations

proof carried
sign-changing bracket

same
root

Newton's method

update clue
local residual + slope

recorded cost
4 evaluations

proof carried
local tangent estimate

Iteration count is a cost clue, not a validation check.

A midpoint is not enough: the bracket must change sign

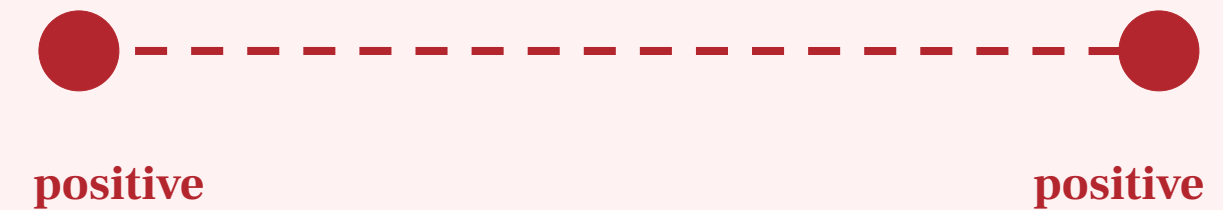
Arithmetic creates a midpoint; physics justifies the bracket

Valid Core bracket



opposite endpoint signs → start bisection

Defective same-sign bracket



do not start bisection

Same-sign endpoints do not prove no root exists; they fail to provide the required bracket.

Transfer the residual pattern to new physics

The physics changes; the reasoning chain stays stable

$$\begin{aligned} \text{thin lens: } f(v) &= \frac{1}{f} - \frac{1}{u} - \frac{1}{v} = 0 & [f(v) \text{ in m}^{-1}, v \text{ in m}] \\ \text{RC charging: } f(t) &= V_s (1 - e^{-t/(RC)}) - V_{\text{threshold}} = 0 & [f(t) \text{ in V}, t \text{ in s}] \end{aligned}$$

Thin lens

root: image distance v in m
residual: inverse-length mismatch in m^{-1}

RC charging

root: threshold time t in s
residual: voltage mismatch in V

model



residual



bracket / start



update



validate

Exit ticket — explain a root to a future you

Close the loop from physical condition to checkable result

1 Define the target

What physical condition must be true?

2 Name the residual

What mismatch is measured, and in what unit?

3 Explain one update

Why is this midpoint or tangent step valid?

4 Validate by substitution

Does the model meet the target, and what is the root unit?

A numerical answer becomes physics when the residual and the units make sense.